

1.3 - StorageCompression - ExamQs - ANSWERS.pdf

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IGCSE Computer Science

1.3 Data Storage & Compression

Exam Style Questions

ANSWERS

Max: 28

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Question 1

Michele wants to email a file to Elsa. The file is too large so it must be compressed.

- a) Name **two** types of compression that Michele could use.

Compression type 1
 Compression type 2

[2]

- b) The file Michele is sending contains the source code for a large computer program. Identify which type of compression would be most suitable for Michele to use. Explain your choice.

Compression type

Explanation

Any three from:

- The file can be restored to the original
- No data can be lost (lossy would remove data)
- ...or will not run correctly
- (Lossless) will give repeating words a value// use RLE
- Value is recorded in an index

[4]

- c) Elsa wants to email some photos to Michele. Many of the photos are very large files, so Elsa needs to reduce their file size as much as possible. Identify which type of compression would be most suitable for Elsa to use. Explain your choice.

Compression type

Explanation

Any three from:

- a) Lossy would reduce the file size more (than lossless)
- b) The redundant data can be removed from the files
- c) Images can still be a similar quality
- d) No requirement for the files to be exactly the same as original file
- e) Photos can be sent quicker // faster to upload / download

[4]

2

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Question 2

Miriam needs to use a large high-resolution photo as a thumbnail image on a website. She will use lossy compression to reduce the file size of the photo to create the thumbnail image.

a) State why a smaller file size is appropriate for this situation.

Smaller file size reduces download / display time // reduces upload time

[1]

b) Explain how lossy compression reduces the file size.

Any four from:

- A compression algorithm is used
- Permanently deleting some data
- Colour depth / palette can be reduced
- Resolution can be reduced
- Fewer bits will be required for each pixel / colour

[4]

Question 3

The following text is stored as a text file:

She sells sea shells on the seashore.
The shells that she sells are sea shells I am sure.

Explain how lossless compression would compress this file.

Five from:

- A (compression) algorithm is used
- No data is removed in the process // original file can be restored
- Repeated words // Patterns in the data (are identified)
 - « and are indexed/put into a table // by example
 - « and are replaced with their index // by example
 - « their positions are stored // by example
 - « the number of times the word/pattern appears is stored // by example

Other valid methods of Lossless compression can be awarded marks

[5]

Question 4

Computer memory size is measured in multiples of bytes.

Four statements about computer memory sizes are given in the table.

Tick (✓) to show if the statement is True or False.

Statement	True (✓)	False (✓)
25kB is larger than 100 MB		✓
999MB is larger than 50GB		✓
3500kB is smaller than 2GB	✓	
2350bytes is smaller than 2kB		✓

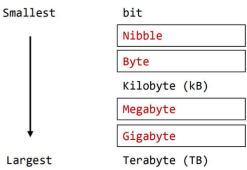
[4]

Question 5

Different units of data can be used to represent the size of a file, as it changes in size.
Fill in the missing units of data, using the list given:

- byte
- gigabyte (GB)
- megabyte (MB)
- nibble

The units of data increase in size from smallest to largest.



[4]